

SMARTPHONE-BASED LIFE-SPACE MOBILITY TRACKING IN OLDER ADULTS UNDERGOING JOINT REPLACEMENT: A FEASIBILITY STUDY

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INTRODUCTION

Life-space mobility describes the extent of a person's movement through their environment and reflects their capacity to take part in meaningful activities [1]. It has historically been assessed by questionnaires, which are subject to recall bias and lack context on how and where people move around their communities. Smartphones passively record location data, but the feasibility and utility of such data for quantifying life-space mobility is unclear. We explored this in older adults undergoing joint replacement, a group in whom life-space mobility is expected to meaningfully improve postoperatively [2].

METHODS

Fifteen adults aged 60 years or older, scheduled for elective hip or knee replacement at Barwon Health, were recruited between August and December 2025. Participants were provided with an Android smartphone for 90 days post-surgery and asked to carry it whenever they left their home. Google Location History data was exported from devices and processed using LS-Hub, a purpose-built platform that removes location data and returns only derived measures: the timing, distance and duration of each journey, the mode of travel, and the Google Places classification for each destination. Feasibility was pre-specified as obtaining valid location data on at least 80% of monitored days. Health-related quality of life (EQ-5D-5L) and joint pain, stiffness and dysfunction (WOMAC) were measured pre-operatively. The validated Life-Space Assessment (LSA) questionnaire was also completed pre-operatively and monthly during the observation period.

RESULTS AND DISCUSSION

All fifteen participants completed baseline questionnaires (Table 1) and 14 (93.3%) provided Google Location History data; one did not return the device. Including this participant, mean±SD data availability was 72.0%±25.5 (median 81.1%, range 0–98.9%). Among participants who returned the device, mean±SD data availability was 77.1%±16.5 (median 82.2%, range 44.4–98.9%) and 8 of 14 participants provided data on ≥80% of monitored days.

A total of 1,947 journeys were recorded (mean 139±62 per participant). Participants travelled a total of 19,363 km over 873 hours of journey duration. Median journey distance was 4.2 km (IQR 1.5–11.4) and 41.5% of journeys were shorter than 3 km. Travel mode was

overwhelmingly by car (95.5%), with walking accounting for 3.5% of journeys. A total of 460 unique journey destinations were recorded, categorised as residential (44.9%), shopping (27.1%), health and wellness (7.6%), entertainment and recreation (6.5%) and food and drink (6.2%). The proportion of journeys to health and wellness destinations fell from 13.2% in weeks 1–4 post-surgery to 5.6% in weeks 9–12.

Journey frequency rose from 0.94±0.64 journeys per day of available data in week 1 post-surgery to a peak of 2.55±1.26 in week 9 and then remained relatively stable. Mean LSA scores followed a similar pattern, falling from 75.5±23.4 pre-operatively to 47.5±24.0 three weeks after surgery, and returning to 76.7±15.4 by week 12.

CONCLUSIONS

Data availability for Google Location History collected via a provided smartphone did not achieve pre-specified feasibility thresholds in older adults recovering from knee and hip replacement surgery. Nevertheless, available data demonstrated similar changes in life-space mobility to the gold-standard LSA questionnaire, and provided additional context on distances travelled, travel modes and the characteristics of places visited. Further research is required to improve the acceptability and feasibility of this method.

REFERENCES

- [1] Rantakokko M, et al. Aging Clin Exp Res. 28: 891-7, 2016.
- [2] Ethgen O, et al. J Bone Joint Surg Am 86: 963-74, 2004.

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Table 1. Pre-operative participant characteristics. Mean ± SD unless stated.

Characteristic	n = 15
Age (years)	69.9 ± 5.1
Female, n (%)	7 (46.7)
Educated beyond secondary, n (%)	6 (40.0)
Retired, n (%)	9 (60.0)
Current/previous smoker, n (%)	11 (73.3)
Chronic diseases, median (range)	3 (0–4)
LSA score (of 120)	75.5 ± 23.4
WOMAC score (of 96)	55.9 ± 11.8
EQ-5D-5L index, median (range)	0.552 (0.029–0.919)